

Python with Generative AI Agent - Full Course Syllabus

Module 1

Python Basics, variables, loops, functions, OOP

Module 2

File handling, exceptions, modules, virtual environments

Module 3

NumPy, Pandas, data processing

Module 4

Generative AI fundamentals, LLMs, tokens, embeddings

Module 5

Prompt Engineering techniques

Module 6

OpenAI API integration

Module 7

LangChain: chains, prompts, memory

Module 8

Vector Databases, RAG, FAISS/Chroma

Module 9

AI Agents, tools, function calling, MCP basics

Module 10

Projects: chatbot, PDF Q&A, AI assistant, automation, deployment

Career Opportunities: GenAI Engineer, AI Agent Developer, Python Developer, LLM Engineer, Prompt Engineer, AI Automation Engineer.